

Homework 1 – Sessions 1 & 2

Algebra, order and justification · Sets, operations and double inclusion

Part A · Session 1 – algebra, order and justification

1 EXPONENTS

Simplify $\frac{16^{3/4} \cdot 2^{-3}}{8^{1/3}}$, naming the exponent law used at each step.

SOLUTION $16^{3/4} = (2^4)^{3/4} = 2^3 = 8$ by $(a^m)^n = a^{mn}$; $2^{-3} = \frac{1}{8}$ by $a^{-n} = 1/a^n$; $8^{1/3} = (2^3)^{1/3} = 2$ by the same power law. So the expression is $\frac{8 \cdot \frac{1}{8}}{2} = \frac{1}{2}$.

The two uses of $(a^m)^n = a^{mn}$ are the fragile steps: that law was stated for $a > 0$. With $a = -1$, $m = 2$, $n = \frac{1}{2}$ it fails, since $((-1)^2)^{1/2} = 1$ but $(-1)^{2 \cdot 1/2} = -1$. Here the base is $2 > 0$, so the steps are legal.

2 MISSING HYPOTHESIS

Each line below is wrong as written. For each, give a counterexample, then either add a hypothesis that makes it true or explain why no such repair exists.

a. $\sqrt{ab} = \sqrt{a} \sqrt{b}$

b. $\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}$

SOLUTION (a) At $a = b = -1$ the left side is $\sqrt{1} = 1$ while the right side is not defined in $\mathbb{R}$. Repair: require $a, b \geq 0$; then both sides are the non-negative number whose square is ab .

(b) At $a = b = 1$ the left side is $\frac{1}{2}$ and the right side is 2 . No hypothesis repairs it into a law – clearing denominators shows the two sides agree only when $(a+b)^2 = ab$, a scattered family of exceptional pairs, not a rule.

3 ABSOLUTE VALUE

Write “ x is at least 5 away from -2 ” as an absolute-value inequality and solve it. Explain why the solution set splits into two pieces.

SOLUTION Distance from -2 is $|x - (-2)| = |x + 2|$, so the sentence is $|x + 2| \geq 5$. $|x + 2| \geq 5 \iff x + 2 \geq 5 \text{ or } x + 2 \leq -5$ (equivalence: reading the definition of $|\cdot|$ by cases) $\iff x \geq 3 \text{ or } x \leq -7$ (equivalence: adding -2 preserves order and is reversible).

The endpoints are included, since “at least 5 away” allows a distance of exactly 5. The set splits because distance ignores direction: a point can be far from -2 on either side of it.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$.

SOLUTION $2x^2 - 12x + 23 = 2(x^2 - 6x) + 23 = 2[(x - 3)^2 - 9] + 23 = 2(x - 3)^2 + 5$. Since $(x - 3)^2 \geq 0$ for every real x , the expression is at least 5, with equality exactly when $x = 3$. So the minimum value is 5, attained at $x = 3$.

5 RATIONAL INEQUALITY

Find all real x with $\frac{2}{x - 1} \leq 1$. Handle the sign of $x - 1$ explicitly; do not multiply through by an expression whose sign you do not know.

SOLUTION First, $x = 1$ is excluded – the left side is undefined there.
Case $x > 1$. Then $x - 1 > 0$, so multiplying preserves the direction: $2 \leq x - 1$, i.e. $x \geq 3$. Combined with $x > 1$ this gives $x \geq 3$.
Case $x < 1$. Then $x - 1 < 0$, so multiplying reverses the direction: $2 \geq x - 1$, i.e. $x \leq 3$, which every $x < 1$ already satisfies. So the whole ray works – as it must, since a negative number is certainly ≤ 1 .
 Solution set: $x < 1$ or $x \geq 3$.

6 QUADRATIC REASONING

Solve

$$2x^2 - 8x + 3 = 0$$

by completing the square. Do not use the quadratic formula.

- Calculate the discriminant and state what it tells you about the number of real solutions.
- Find the roots by completing the square, writing each line as an equivalence.

SOLUTION (a) Here $a = 2$, $b = -8$, and $c = 3$, so

$$\Delta = b^2 - 4ac = (-8)^2 - 4(2)(3) = 40 > 0.$$

Therefore the equation has two distinct real roots.

(b)

$$\begin{aligned} 2x^2 - 8x + 3 = 0 &\iff x^2 - 4x + \frac{3}{2} = 0 \iff x^2 - 4x = -\frac{3}{2} \\ &\iff (x - 2)^2 - 4 = -\frac{3}{2} \iff (x - 2)^2 = \frac{5}{2}. \end{aligned}$$

Therefore,

$$x - 2 = \pm \sqrt{\frac{5}{2}}, \quad x = 2 \pm \sqrt{\frac{5}{2}} = 2 \pm \frac{\sqrt{10}}{2}.$$

7 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it: $1 \in C$, $\{1\} \subseteq C$, $\{1, 2\} \subseteq C$, $\emptyset \subseteq C$.

SOLUTION $1 \in C$: true – 1 is listed as an element. $\{1\} \subseteq C$: true – the only element of $\{1\}$ is 1, and $1 \in C$, which is what $\subseteq$ demands. $\{1, 2\} \subseteq C$: false – it would need $2 \in C$, and 2 is not an element of C (the set $\{1, 2\}$ is, which is a different statement). $\emptyset \subseteq C$: true – vacuously, for every set: $\emptyset$ has no element that could fail to lie in C .

8 OPERATIONS

Let $U = \{1, 2, \dots, 15\}$, $A = \{x \in U : x \text{ is a multiple of } 3\}$ and $B = \{x \in U : x \text{ is odd}\}$. Write A and B in roster notation, compute $A \triangle B$ and $(A \cup B)^c$, then describe $(A \cup B)^c$ in plain English and name the law that turns “not (a multiple of 3 or odd)” into a statement with “and” in it.

SOLUTION $A = \{3, 6, 9, 12, 15\}$ and $B = \{1, 3, 5, 7, 9, 11, 13, 15\}$.
 $A \setminus B = \{6, 12\}$ and $B \setminus A = \{1, 5, 7, 11, 13\}$, so $A \triangle B = \{1, 5, 6, 7, 11, 12, 13\}$.
 Since $A \cup B = \{1, 3, 5, 6, 7, 9, 11, 12, 13, 15\}$, the complement is $(A \cup B)^c = \{2, 4, 8, 10, 14\}$.
 In words: the numbers in U that are even and not a multiple of 3. The rewriting is De Morgan's law $(A \cup B)^c = A^c \cap B^c$ – negating an “or” produces an “and”.

9 POWER SET

- With $A = \{a, b\}$ and $B = \{b, c\}$, list $\mathcal{P}(A \cap B)$ and $\mathcal{P}(A) \cap \mathcal{P}(B)$ in full, and check they agree.
- Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ for *all* sets A, B , by double inclusion.

SOLUTION (a) $A \cap B = \{b\}$, so $\mathcal{P}(A \cap B) = \{\emptyset, \{b\}\}$. Also $\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$ and $\mathcal{P}(B) = \{\emptyset, \{b\}, \{c\}, \{b, c\}\}$, whose intersection is $\{\emptyset, \{b\}\}$ – the same set.
 (b) ($\subseteq$) Let $X \in \mathcal{P}(A \cap B)$ be arbitrary, so $X \subseteq A \cap B$. Every element of X then lies in A and in B , so $X \subseteq A$ and $X \subseteq B$, i.e. $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$.
 ($\supseteq$) Let $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$ be arbitrary, so $X \subseteq A$ and $X \subseteq B$. Let $x \in X$ be arbitrary; then $x \in A$ and $x \in B$, so $x \in A \cap B$. Hence $X \subseteq A \cap B$, i.e. $X \in \mathcal{P}(A \cap B)$. ■

10

DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$, by double inclusion. The word **arbitrary** must appear in each direction.

SOLUTION ($\subseteq$) Let $x \in (A \cap B)^c$ be arbitrary. Then $x \in U$ and $x \notin A \cap B$, i.e. it is not the case that both $x \in A$ and $x \in B$. So $x \notin A$ or $x \notin B$, giving $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$.

($\supseteq$) Let $x \in A^c \cup B^c$ be arbitrary. Then $x \notin A$ or $x \notin B$, so it is not the case that x lies in both, i.e. $x \notin A \cap B$. As $x \in U$, $x \in (A \cap B)^c$. ■

11

DISPROVE, THEN REPAIR

- Disprove: $(A \cup B) \setminus B = A$ for all sets A, B . Name your sets, compute both sides, and point at an element that separates them.
- State the extra condition on A and B that makes the identity true, and prove it under that condition.

SOLUTION (a) Take $A = B = \{1\}$. Then $A \cup B = \{1\}$ and $(A \cup B) \setminus B = \emptyset$, while $A = \{1\}$. The element 1 is in the right side and not the left, so the claim fails.

(b) The condition is $A \cap B = \emptyset$. Assume it. ($\subseteq$) Let $x \in (A \cup B) \setminus B$ be arbitrary: then $x \in A \cup B$ and $x \notin B$, so $x \in A$. ($\supseteq$) Let $x \in A$ be arbitrary: then $x \in A \cup B$, and $x \notin B$, since $A \cap B = \emptyset$ leaves no element in both. So $x \in (A \cup B) \setminus B$. ■

A campus card system records, for one week, which of the **200** students used the gym and which used the library. Take U to be the set of all **200** students, $G \subseteq U$ those who used the gym and $L \subseteq U$ those who used the library. The logs give

$$|G| = 120, \quad |L| = 95, \quad |G \cap L| = 60.$$

- The two sets cut U into four regions. Name each one in set notation and give its size, explaining where each number comes from.
- Every visit is logged as an ordered pair (student, facility), the facility drawn from $F = \{\text{gym, library, pool}\}$. How many such pairs are possible in principle? Name the theorem that gives the count, and say why the pairs must be ordered.

SOLUTION (a) $G \cap L$: both, **60** students, given. $G \setminus L$: gym only, $120 - 60 = 60$, since the **120** gym users split into those who also used the library and those who did not. $L \setminus G$: library only, $95 - 60 = 35$, by the same argument. $(G \cup L)^c$: neither, $200 - (60 + 60 + 35) = 45$, because the four regions are disjoint and every student lies in exactly one of them.

(b) The possible pairs are the elements of $U \times F$, and $|U \times F| = |U| \cdot |F| = 200 \cdot 3 = 600$, by the theorem that a finite Cartesian product has size the product of the sizes — two independent choices multiply. The pairs must be ordered because the coordinates play different roles: $(a, b) = (c, d)$ only when $a = c$ and $b = d$, so (student, facility) is a different record from (facility, student).